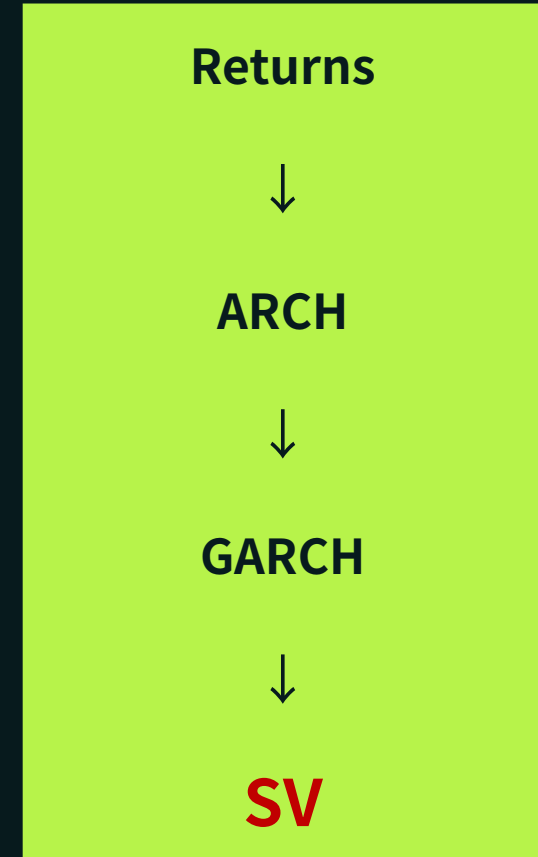


Hand-Calculation Workbook

ARCH and GARCH volatility in five worked problems



Five problems retrace the main deck's calculations in order

Units — returns in %, variance in %², annualization factor $\sqrt{252} \approx 15.87$

W1	————	Returns, annualizing	Getting units right Log return · $\sqrt{252}$
W2	————	ARCH(1) update	Today's variance One big shock vs. a calm day
W3	————	GARCH(1,1) update	Floor, news and memory Three terms in one sum
W4	————	Long run, half-life	Mean reversion $V_L \cdot \text{half-life}$
W5	————	SV step	Uncertainty A range, not a number

Read each problem as: given → solution → answer

W1. 1% daily volatility is 15.9% annual volatility

A price rises from 100 to 102. With 1% daily volatility, what is annual volatility?

Price yesterday

100

Price today

102

Daily vol

1%

$\sqrt{252}$

≈ 15.87

Solution

$$r = \ln \frac{102}{100} = \ln 1.02 \approx 0.0198 \text{ (1.98\%)}$$

$$\sigma_{\text{annual}} = 1\% \times \sqrt{252} \approx 1\% \times 15.87 = 15.9\%$$

Answer: return $\approx 1.98\%$, annual vol $\approx 15.9\%$ — variance scales by 252, std. dev. by $\sqrt{252}$

W2. In ARCH(1), a big shock lifts today's variance over tenfold

$\alpha_0 = 0.2$, $\alpha_1 = 0.5$. Today's volatility after a -3% day versus a $+0.5\%$ day?

α_0
0.2

α_1
0.5

Shock day r
 -3%

Calm day r
 $+0.5\%$

Solution — (A) shock day · (B) calm day · (C) long-run

$$(A) \quad \sigma_t^2 = 0.2 + 0.5 \times (-3)^2 = 0.2 + 4.5 = 4.7, \quad \sigma_t = \sqrt{4.7} \approx 2.17\%$$

$$(B) \quad \sigma_t^2 = 0.2 + 0.5 \times (0.5)^2 = 0.2 + 0.125 = 0.325, \quad \sigma_t \approx 0.57\%$$

$$(C) \quad \bar{\sigma}^2 = \frac{0.2}{1 - 0.5} = 0.4, \quad \bar{\sigma} \approx 0.63\%$$

Answer: 2.17% vs. 0.57% — only the size (square) matters, not the sign

W3. A GARCH(1,1) update is a single sum of three terms

$\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$. Yesterday's variance 1.0 (%²), shock -3%. Today's vol?

 ω **0.02** α **0.10** β **0.88** σ^2 / ε yesterday**1.0 / -3**

Solution

$$\begin{aligned}\sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 = 0.02 + 0.10 \times 9 + 0.88 \times 1.0 \\ &= 0.02 + 0.90 + 0.88 = 1.80\end{aligned}$$

$$\sigma_t = \sqrt{1.80} \approx 1.34\% \Rightarrow 1.34\% \times 15.87 \approx 21.3\%$$

Answer: 1.34% daily, $\approx$ 21.3% annual — news (0.90) and memory (0.88) weigh about equally

W4. Same parameters: 15.9% long-run vol, half-life of about 34 days

Using W3's parameters, find the long-run variance V_L and the shock half-life

 ω **0.02** $\alpha + \beta$ **0.98** $1 - \alpha - \beta$ **0.02** $\ln 0.5$ **-0.693**

Solution

$$V_L = \frac{0.02}{1 - 0.10 - 0.88} = \frac{0.02}{0.02} = 1.0 \Rightarrow 1\% \text{ daily} \approx 15.9\% \text{ annual}$$

$$\text{half-life} = \frac{\ln 0.5}{\ln 0.98} = \frac{-0.693}{-0.0202} \approx 34 \text{ days}$$

Answer: $V_L = 1.0$ (%²), half-life ≈ 34 days — today's 21.3% drifts back to 15.9%

W5. In SV, today's volatility remains a range even given yesterday

$\mu_h = 0$, $\phi = 0.95$, yesterday's $h = 0.4$, s.d. of $\eta = 0.2$ — today's vol $\sigma_t = e^{(h_t/2)}$

μ_h

0

ϕ

0.95

h yesterday

0.4

s.d. of η

0.2

Solution

$$E[h_t] = 0 + 0.95 \times (0.4 - 0) = 0.38 \Rightarrow \sigma_t = e^{0.38/2} = e^{0.19} \approx 1.21\%$$

$$h_t = 0.38 \pm 0.20 \Rightarrow \sigma_t \in [e^{0.09}, e^{0.29}] \approx [1.09\%, 1.34\%]$$

Answer: center 1.21%, ± 1 s.d. range 1.09–1.34% — GARCH gives a number, SV a distribution

**Add variances,
take square roots for volatility,
and scale by $\sqrt{252}$ to annualize.**

**Back to the main deck
Re-read with the charts**

Parameters are illustrative